T28 FCBGA pad cratering의 열사이클 손상

Minho Oh · KAIST

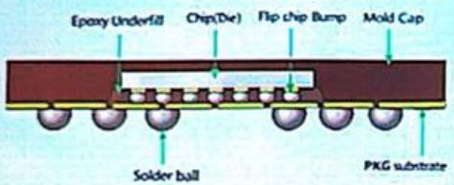
Introduction

Research background

FCBGA-Based semiconductor packaging

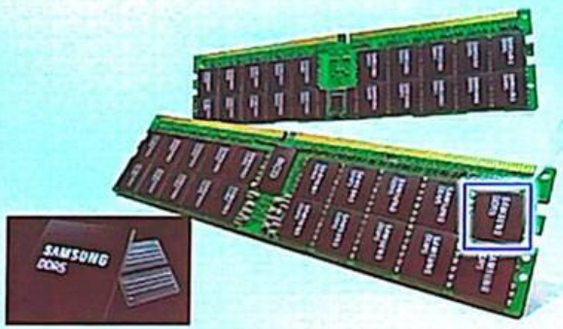


Samsung-electro mechanics

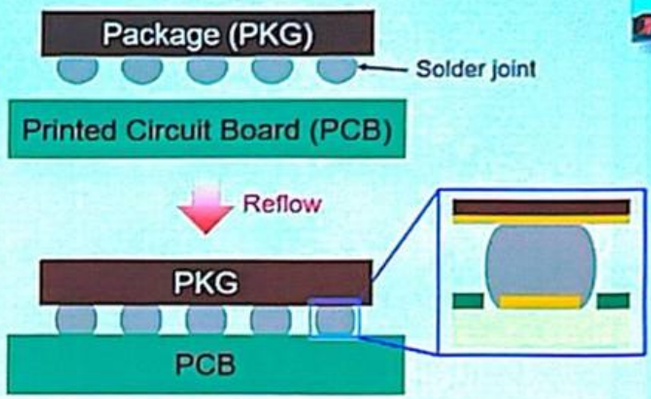


Daeduck electronics

Advanced DRAM module

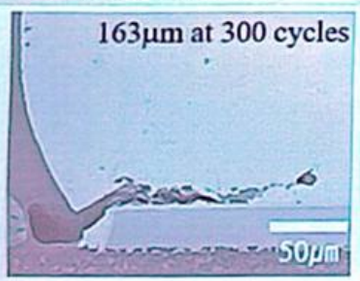


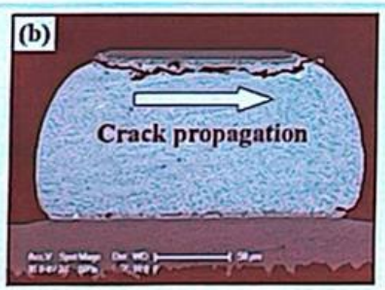
Samsung electronics



Mechanical reliability issue

Solder joint crack

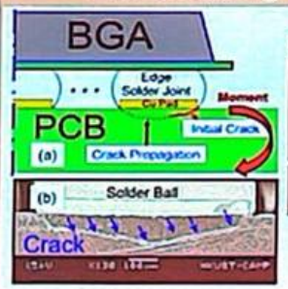


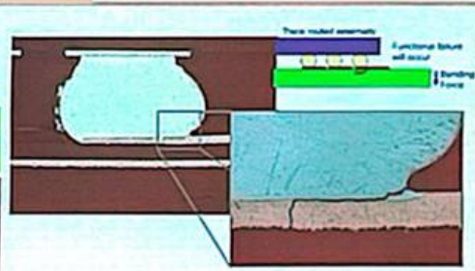


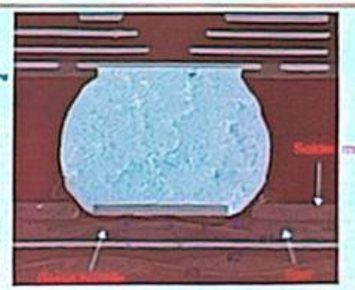
- Failure between solder and IMC interface
- Conventionally observed failure modes.

Park, G. J., et al., *Engineering Failure Analysis, 157*, 2024.
Kim, D. G., et al., *Thin solid films, 504* (1-2), 2006.

Pad cratering







- Failure occurs at the pad beneath the solder joints.
- High-strength solder concentrates the load on the pad[1-2] → **Increasingly reported**

Zhang, Q., et al., *IEEE 67th ECTC, 2017* / Hetnarski, R., Springer Netherlands, 2014
[1] Chong, D. Y., et al., Microelectronics reliability, 46(7), 2006
[2] Loukia, M., et al., IEEE 20th iTherm, 2021

4

T28 FCBGA pad cratering의 열사이클 손상

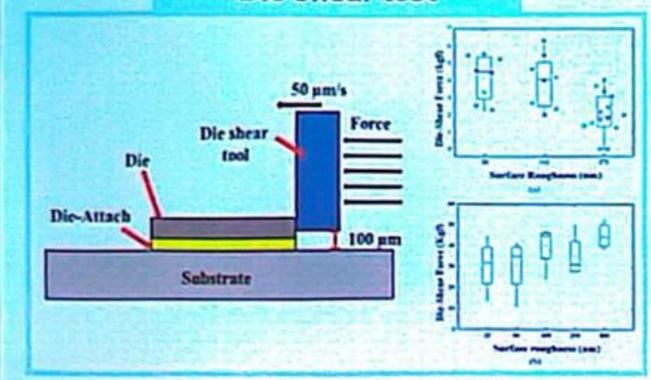
Minho Oh · KAIST

Introduction

Research background

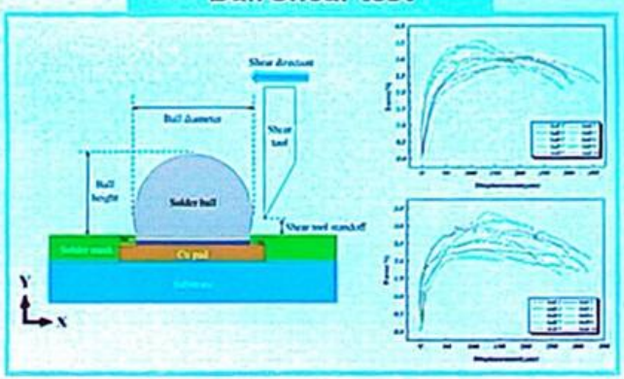
Test Methods for Joint Reliability

Die shear test



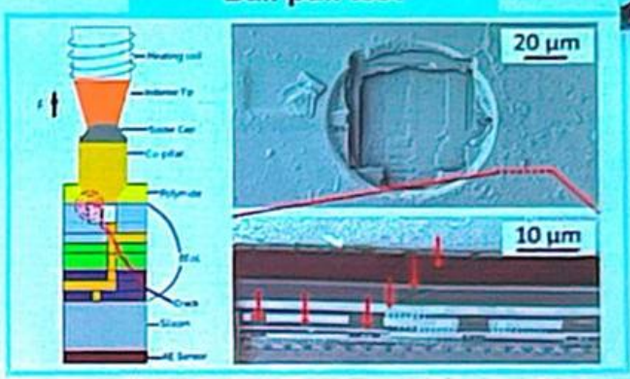
Navarro, L. A., et al., *IEEE TPEL*, *29*(5), 2013

Ball shear test



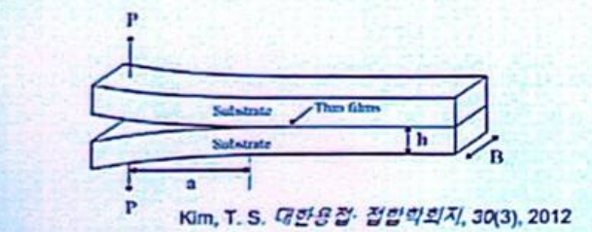
Liu, L., Yu, et al., *Engineering Fracture Mechanics*, *316*, 2025

Ball pull test

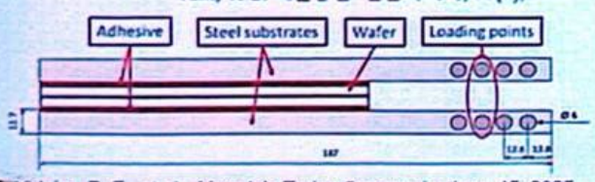


Silomon, J., et al., *Microelectronics Reliability*, *136*, 2022

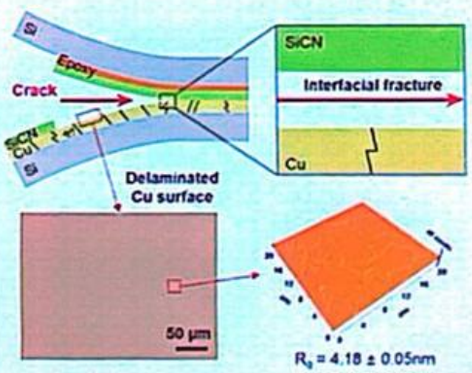
Double cantilever beam (DCB) test



Kim, T. S. *대한용접·접합학회지*, 30(3), 2012



Pereira, P. F., et al., *Materials Today Communications*, 45, 2025

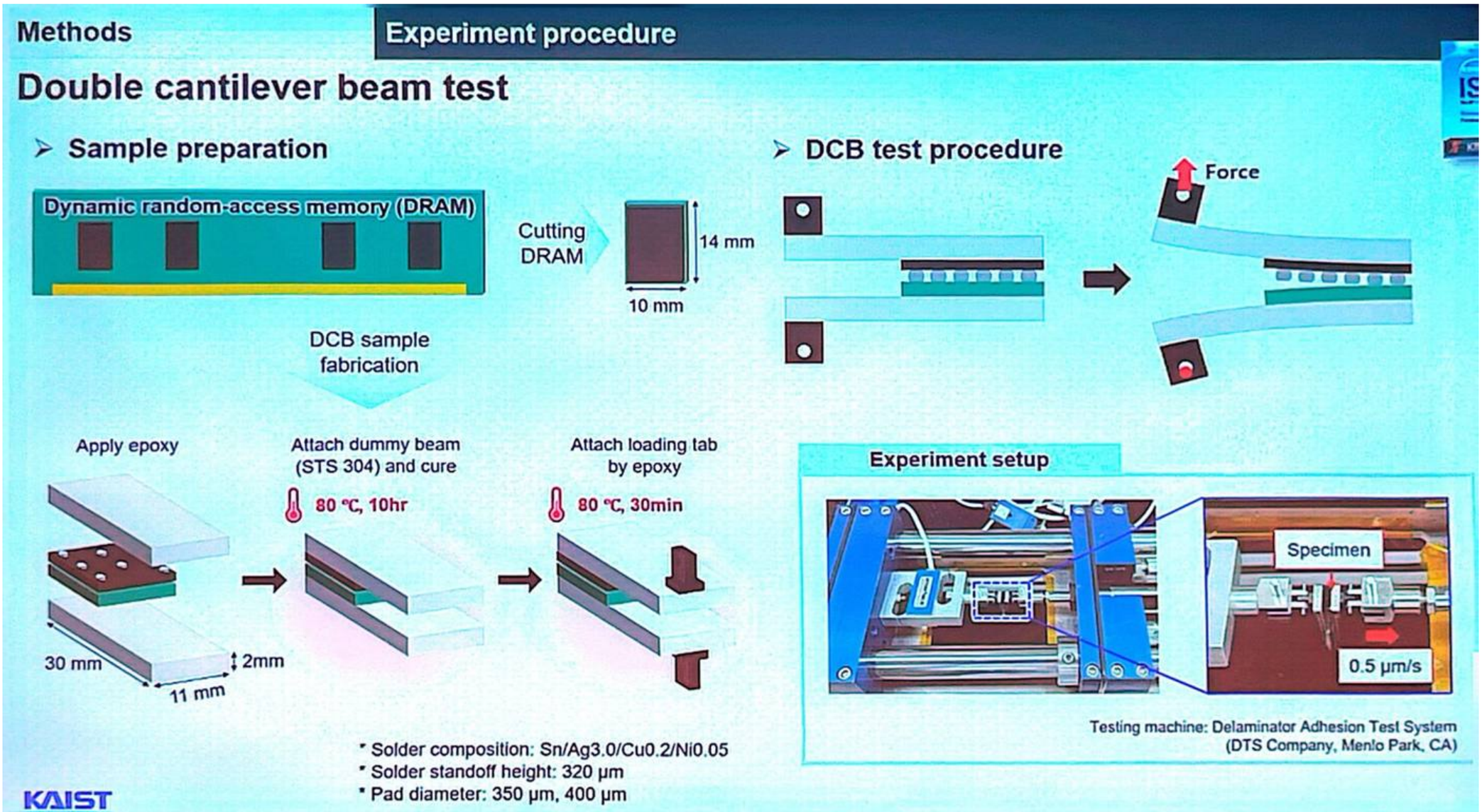


Kim, T. S., et al., *Applied Surface Science*, *688*, 2025

- **Identification of the weak interface**
- **Failure analysis based on fracture mechanics**
  - Quantitative measure of strain energy release rate

T28 FCBGA pad cratering의 열사이클 손상

Minho Oh · KAIST



T28 FCBGA pad cratering의 열사이클 손상

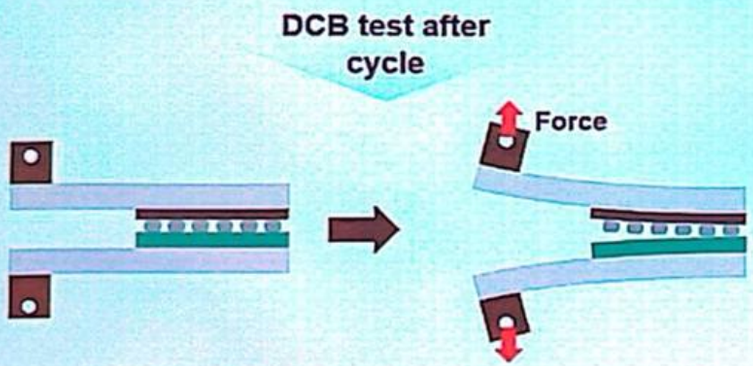
Minho Oh · KAIST

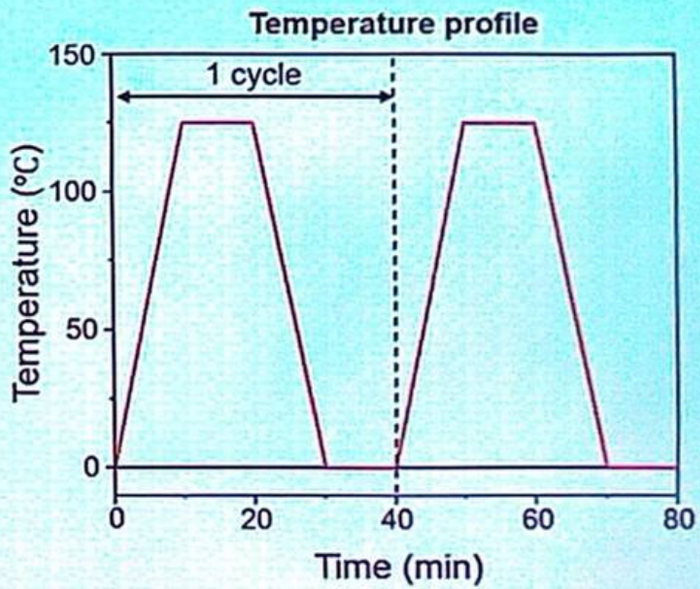
thods

ermal cycle test

Experiment procedure







- To quantitatively evaluate the change in adhesion, thermal cycling was applied to the DRAM module.
  - 0–125 °C, 10 min dwell, 10 min ramp (following JEDEC standard)[1]
- DCB tests were conducted **after 0, 200, 400, and 500 cycles.**

ST

[1] JESD22-A104F.01

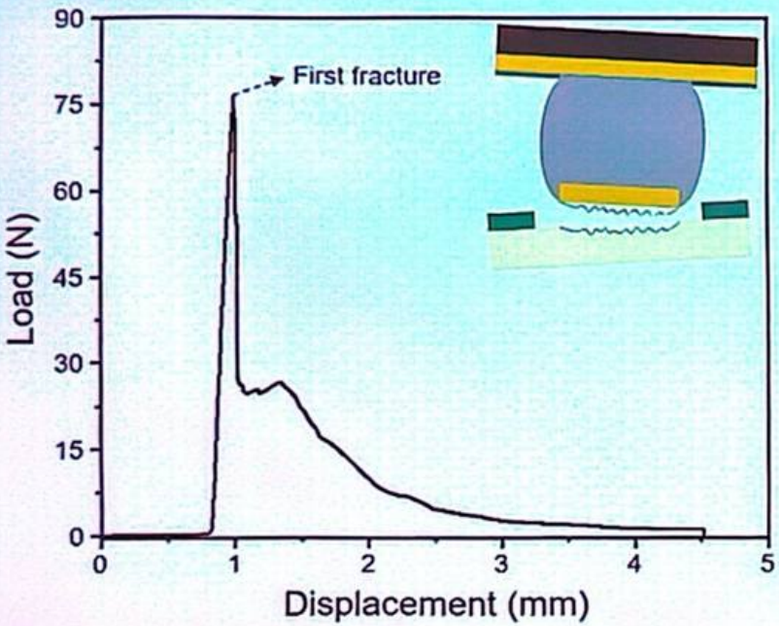
T28 FCBGA pad cratering의 열사이클 손상

Minho Oh · KAIST

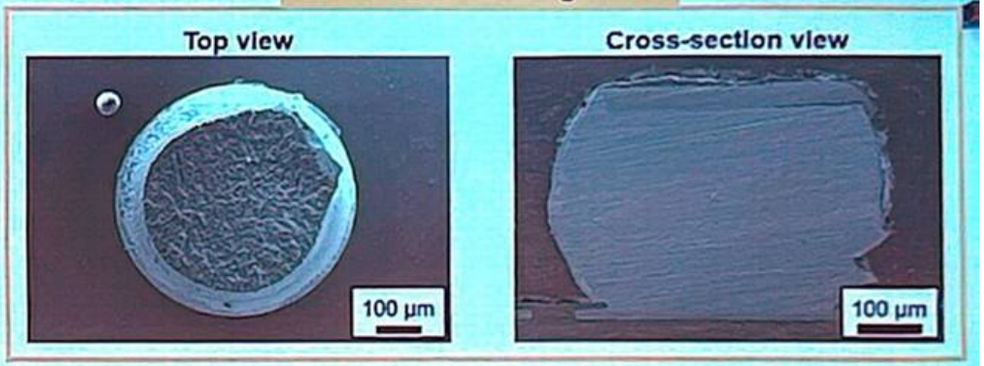
Results and discussion

Experiment results

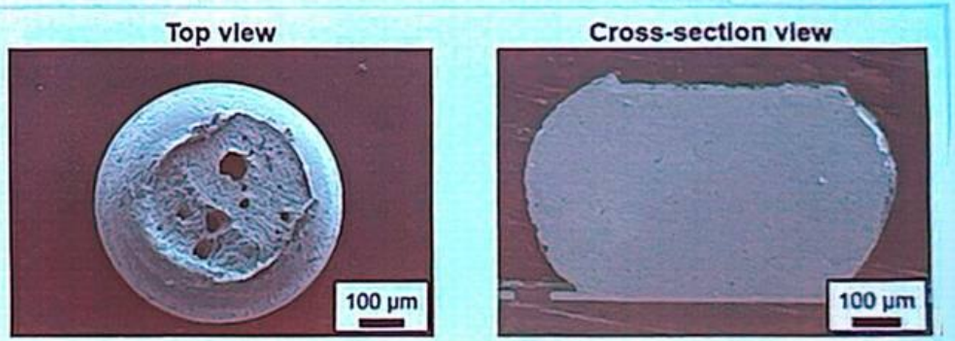
Double cantilever beam test results



Pad cratering



Solder cohesive fracture



- Load increased until first-row delamination, followed by a sharp drop.
- **Pad cratering dominant (80%~90%)**; solder cohesive fracture at limited locations.

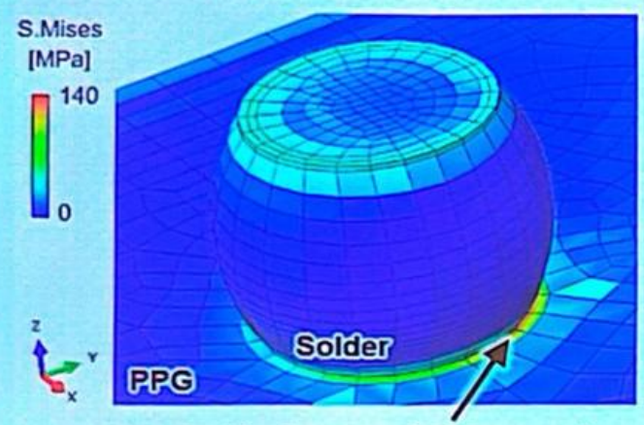
T28 FCBGA pad cratering의 열사이클 손상

Minho Oh · KAIST

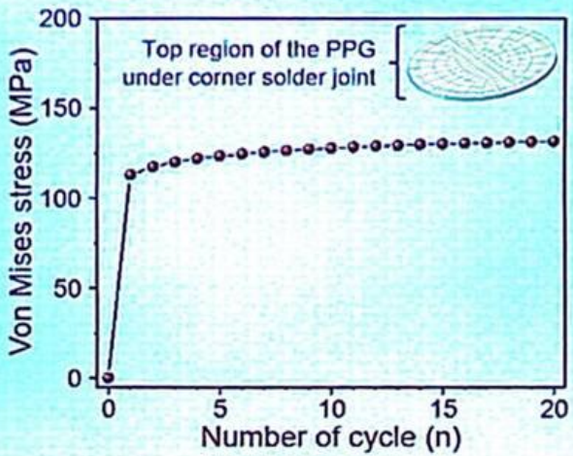
Results and discussion | Thermal cycle test simulation results

Accumulation of residual stress

➢ Accumulation of residual stress

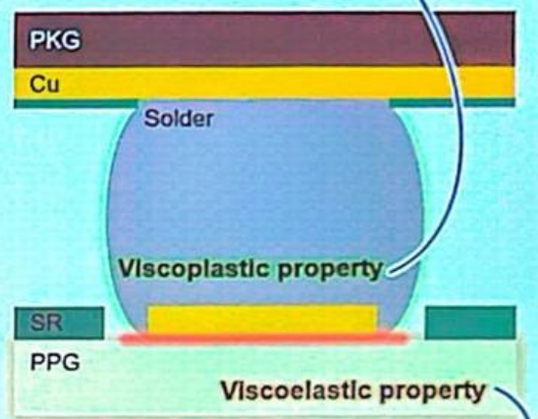


Residual stress of PPG around solder joint



➢ Material characteristic

Anand [1, 2]: $\dot{\epsilon}_p = A exp\left(-\frac{Q}{RT}\right)\left[\sinh\left(\xi\frac{\sigma}{s}\right)\right]^{\frac{1}{m}}$



Prony [3]: $E(t) = E_\infty + \sum_{i=1}^{n} E_i \exp\left(-\frac{t}{\tau_i}\right)$

WLF [4]: $log a_T = -\frac{C_1(T-T_0)}{C_2+(T-T_0)}$

- **Residual stress accumulated** over the thermal cycles in the simulation.
- This is attributed to the time- and temperature-dependent material properties, which prevent full stress recovery upon returning to room temperature.

[1] Anand, L. *International Journal of plasticity*, *1*(3), 1985
[2] Brown, S. B., et al., *International Journal of plasticity*, *5*(2), 1989
[3] Leng, Z., et al., *Computer-aided civil and infrastructure engineering*, *36*(11), 2021
[4] Dagdug, L., et al., *Physica A: Statistical Mechanics and its Applications*, *250*(1-4), 1998

KAIST 19